

Cracking the Evaluation Paradox: Building AI Quality Assessment Systems That Actually Work

Why high BLEU/ROUGE scores can hide dangerous errors, and a hybrid framework that catches them

1. Scenario and Method

A summarization product launched with strong benchmark scores, but users reported wrong figures, missing details and invented information. I computed BLEU and ROUGE for three model versions (V1, V2, V3) on 10 articles (business 2, technology 3, health 3, science 2), read all 30 summaries against their source articles, catalogued failures, and designed a hybrid framework. A script also checks every number in a summary against the source article.

2. Computed Metrics (Deliverable 1)

Model	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore F1
V1	0.0493	0.4026	0.1281	0.3152	pending
V2	0.4878	0.7404	0.5553	0.6879	pending
V3	0.1202	0.4835	0.1899	0.3686	pending

Highest per metric: V2 on BLEU, ROUGE-1, ROUGE-2 and ROUGE-L. **BERTScore:** not computed in my environment (the model download was unreachable); `compute_bertscore.py` produces it locally.

BLEU, ROUGE-1, ROUGE-2 and ROUGE-L agree completely: V2 ranks first, V3 second and V1 last. V2 leads by a wide margin (BLEU 0.488 against 0.120 for V3 and 0.049 for V1), so on this dataset the metrics do identify the best summarizer. The disagreement is in what the ranking implies. V3 beats V1 on every metric, yet my manual review found factual errors in six of V3's ten summaries (wrong revenue, wrong dates, a false claim about stolen Social Security numbers) and none in V1, whose problem is vagueness. Within V3, summaries with errors score almost the same ROUGE-L as correct ones (0.358 against 0.385). BLEU counts exact n-gram overlap and is harshest on paraphrase; ROUGE measures how much reference content is recalled; BERTScore compares contextual embeddings and should reward paraphrases that BLEU punishes. None of them checks facts against the source. BERTScore must be run locally with `compute_bertscore.py`; its ranking should be added before submission.

3. Manual Quality Assessment (Deliverable 2): all 30 summaries

E = factual error, M = missing key info, C = coherence problem, N = unnatural language, Q = overall quality (1-5). Full notes are in manual_quality_assessment.csv.

Art	Model	E	M	C	N	Q	R-L	Bad nums	Key issue
1	V1	-	Y	-	-	3	0.44	-	All stated facts correct but vague: omits 31% cloud growth, the \$8B amount, and exceeding expectations.
1	V2	-	Y	-	-	4	0.43	-	Accurate and well written; only omits the \$8B data-centre investment.
1	V3	Y	Y	-	-	2	0.53	52	Revenue stated as \$52B; source says \$45.2B. Fluent and confident, so the error is easy to miss.
2	V1	-	Y	-	-	2	0.30	-	Correct but generic: no 34%, no 2,400 patients, no HbA1c result, no Stanford.
2	V2	-	-	-	-	5	0.72	-	Accurate; captures Phase 3, 34%, 18 months, 2,400 patients, 1.8% HbA1c.
2	V3	Y	Y	-	-	1	0.20	2024	Wrong mechanism (source: liver glucose production / insulin sensitivity) and wrong date (2024; source says lat...
3	V1	-	Y	-	-	2	0.35	-	Correct but omits 48% growth, year, 320-mile range, and 58% share.
3	V2	-	-	-	-	5	0.67	-	Accurate: 342,000 units, 48%, \$7,500 incentives, 320-mile range.
3	V3	-	Y	-	-	4	0.48	-	No factual errors; omits 320-mile range and 58% Tesla share. Good example of a V3 summary that is fine.
4	V1	-	Y	-	Y	2	0.44	-	No numbers at all; 'provisions for coal plants and renewable energy goals' is awkward and uninformative.
4	V2	-	-	-	-	5	0.59	-	Accurate: 55% by 2035, 60% for developed nations, \$300B, coal 2032, 75% renewables.
4	V3	Y	-	-	-	1	0.37	2030;65	Three errors: 65% (source 55%), coal closures by 2030 (source 2032), developed nations net-zero by 2035 (sourc...
5	V1	-	Y	-	-	2	0.35	-	Correct but omits \$500M, \$4.2B valuation, revenue and growth figures.
5	V2	-	-	-	-	5	0.89	-	Accurate and complete; matches source figures.
5	V3	-	Y	-	-	4	0.38	-	Facts verified; omits the European/Asian expansion plans and lists only one lead investor (Sequoia; Tiger Glob...
6	V1	-	Y	-	-	2	0.35	-	Correct but names no compound, institution, or figures.
6	V2	-	-	-	-	5	0.75	-	Accurate: Resolvin-K, MRSA, 99.7% in 48 hours, human trials 2025.
6	V3	Y	Y	-	-	1	0.25	100;2026	Dangerous: claims 100% effectiveness in human trials (none have happened; source: mouse models), pharmacies by...
7	V1	-	Y	-	Y	2	0.29	-	No numbers; 'many knowledge workers work remotely now compared to before' is awkward.
7	V2	-	-	-	-	5	0.76	-	Accurate: 68%, 42% fully remote, 26% hybrid, 78% satisfaction, 18.5% vacancy.
7	V3	-	Y	-	-	4	0.29	-	Facts verified; omits mixed productivity findings and 18.5% office vacancy.
8	V1	-	Y	-	-	2	0.18	-	Correct but no figures (\$0.03/kWh, 89%, 1,200 GW).
8	V2	-	-	-	-	5	0.80	-	Accurate: \$0.03, 89% drop, 1,200 GW, doubling by 2028.
8	V3	Y	-	-	-	1	0.47	28	Panel efficiency given as 28% (source 24%) and capacity 'triple by 2028' (source: double).
9	V1	-	Y	-	-	2	0.28	-	Correct but omits 8.4M customers, data types, and cause (third-party software).
9	V2	-	-	-	-	5	0.57	-	Accurate: 8.4M customers, names/emails/phones, third-party vulnerability, credit monitoring, investigation.

Art	Model	E	M	C	N	Q	R-L	Bad nums	Key issue
9	V3	Y	Y	-	-	1	0.33	-	States SSNs and banking credentials were accessed; the source says they were NOT. Fluent and serious misinform...
10	V1	-	Y	-	-	2	0.16	-	Correct but vague: no 16 children, SCID, 100%, or timeline.
10	V2	-	-	-	-	5	0.70	-	Accurate: 16 children, SCID, 100% vs 75%, six months.
10	V3	-	Y	-	-	4	0.39	-	Facts verified (16 children, 100%, late 2025); omits comparison with bone marrow transplant and viral delivery...

Summary: V1 had 0 factual errors but omitted key information in 10 of 10 (average quality 2.1). V2 had 0 errors (average 4.9). V3 had 6 factual errors in 10 summaries (average 2.3). Human judgment and metrics agree that V2 is best, but metrics rank V3 above V1.

4. Metric Failure Pattern Analysis (Deliverable 3)

Automated metrics consistently miss four kinds of problems: wrong numbers and dates, fabricated or overstated claims, false statements about what the source says (including negations), and the harm done by vagueness versus the harm done by falsehood. BLEU, ROUGE and BERTScore all compare a summary to a reference summary, never to the source article, and none of them has any way to know whether a claim is true.

The data gives clear examples. First, article 1, V3 reports revenue of \$52 billion; the source says \$45.2 billion. Its ROUGE-L is 0.535, higher than V2's correct summary (0.429). Second, article 8, V3 gives panel efficiency as 28% (source: 24%) and says capacity will triple by 2028 (source: double), yet scores ROUGE-L 0.465 against 0.177 for V1, which is accurate but generic. Third, article 9, V3 says hackers accessed Social Security numbers and banking credentials, while the source states they were not accessed. This is a serious error with ROUGE-L 0.333, above V1's correct 0.282. Fourth, article 6, V3 claims 100% effectiveness in human trials when only mouse studies exist, and scores 0.247 only because it omits other content.

Across the dataset, V3 averages BLEU 0.120 and ROUGE-L 0.369, beating V1 (0.049 and 0.315), though V3 has six factually wrong summaries and V1 has none. Inside V3, wrong and correct summaries score alike (ROUGE-L 0.358 versus 0.385).

The reasons are structural. A changed number such as 52 for 45.2 alters one token in about forty, so n-gram overlap barely moves. Embedding similarity, which BERTScore uses, treats numerals and negations as near neighbours in meaning. Metrics reward fluency and topical match, and penalise omission more than falsehood, which inverts the real risk: an unhelpful summary is safe, a confident wrong one is dangerous. Quality dimensions such as factual accuracy, truthfulness and risk need source-grounded checks and human review.

5. Hybrid Evaluation Framework (Deliverable 4)

Component	Design
Weights	BLEU 10%, ROUGE-L 30%, BERTScore F1 60% (total 100%)
Review triggers	ROUGE-L < 0.35; BERTScore F1 < 0.80; composite < 0.60; high-risk keyword; any flag in health/finance/security
Auto-reject	Any number not found in the source article
Auto-approve	No rule violation and all thresholds met
Reviewer checklist	Factual accuracy, hallucination, negation, completeness, coherence, naturalness/tone
Target review rate	15-25% of production traffic, tuned monthly

Test on the 30 lab summaries: 15 approved, 10 pending review, 5 rejected. All 6 factual errors were rejected or flagged; none were approved. The best model (V2) had a 10% review rate. This test used the rules and ROUGE-L only because BERTScore was not available.

Rationale (232 words)

Weights reflect what the failure analysis showed. BLEU receives only 10% because it is the harshest on paraphrase and had the smallest practical value: V3 and V1 were both near zero. ROUGE-L gets 30% as a coverage signal, useful for spotting omissions such as V1's vague summaries. BERTScore gets 60% because it captures meaning rather than exact wording, though it must be recalibrated on local results because it was not computed in this analysis. Since no metric detected the six factual errors in V3, the composite score never decides on its own; it only ranks and flags. The decisive controls are rules and humans. Any number in a summary that does not appear in the source automatically rejects it, which caught five of six V3 errors. A keyword trigger for high-stakes claims (Social Security, banking credentials, cure, 100%, availability dates) sends the sixth to review, because an invented security claim contains no new number. Low ROUGE-L below 0.35 triggers review for missing content. On the 30 test summaries every factual error was rejected or flagged and none were approved. Review load on the best model (V2) was 10%, while the full

mixed test set reached 33% because it deliberately contains weak outputs; the target in production is 15-25%, tuned monthly. Reviewers follow a fixed checklist, with factuality first. Thresholds should be recalibrated against reviewer decisions every month and whenever the model changes.

6. Files

computed_metrics_analysis.md, manual_quality_assessment.csv, metric_failure_analysis.md, hybrid_evaluation_framework.json, plus compute_metrics.py, compute_bertscore.py and build_deliverables.py for reproducibility.